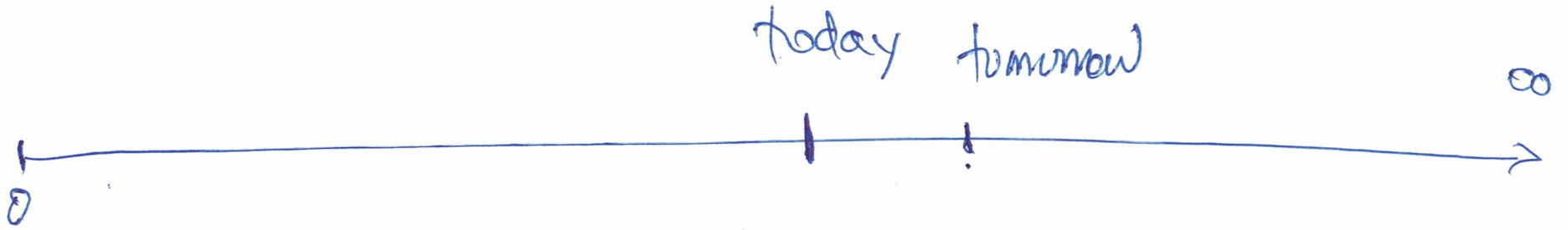
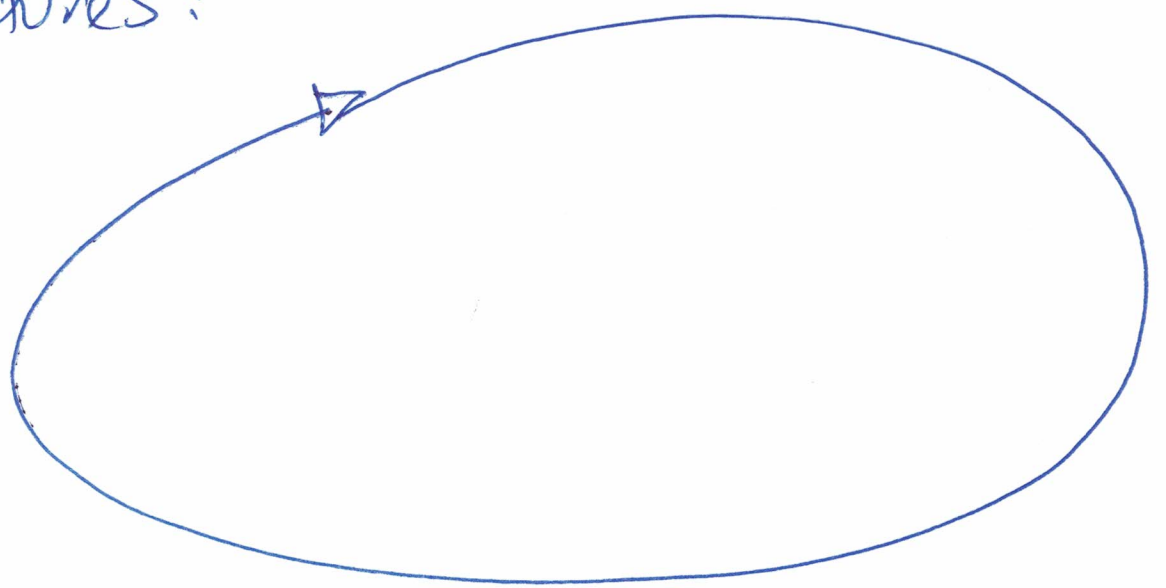
Cpts 317. 9/9.

math induction:



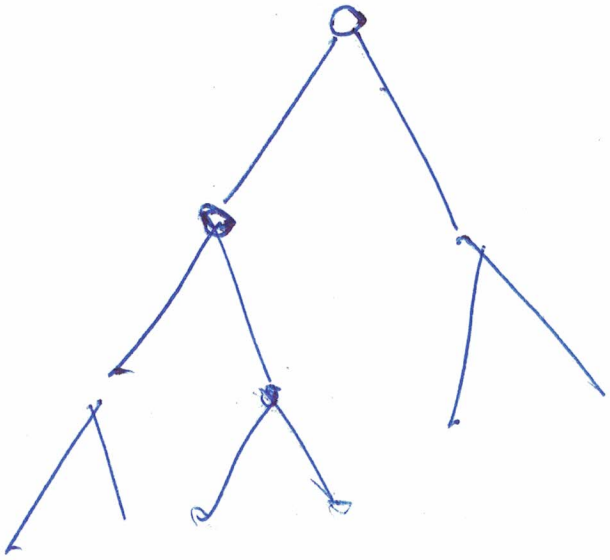
(structure is a line)

other structures:



movie

"arrival"



Recall def's:

1. $\emptyset$ is reg
2. $\{\Lambda\}$ is reg
3. $\{a\}$, $a \in \Sigma$, is reg
4. if L_1 and L_2 are reg, then so are

(4.1) $L_1 + L_2$

(4.2) $L_1 \cdot L_2$

(4.3) L_1^*.

Exaple. Using structural Induction to prove:

$Appear(L, b) = \{x : x \in L$ and x contains symbol $b\}$

is reg if L is reg.

// Exaple. $L = \{abbba, ab, aaa\}$

// $Appear(L, b) = \{abbba, ab\}$

Proof. use structural Induction. According to def of reg languges, we have:

1. $L = \emptyset$. $Appear(L, b) = \emptyset$ is reg.

2. $L = \{\lambda\}$. $Appear(L, b) = \emptyset$ is reg

3. $L = \{a\}$, $a \in \Sigma$. Then

$$Appear(L, b) = \begin{cases} \{b\} & b = a \\ \emptyset & \text{o.w.} \end{cases}$$

is reg,

I.H.

4. if L_1 and L_2 are reg, and $Appear(L_1, b)$ and $Appear(L_2, b)$ are reg, then

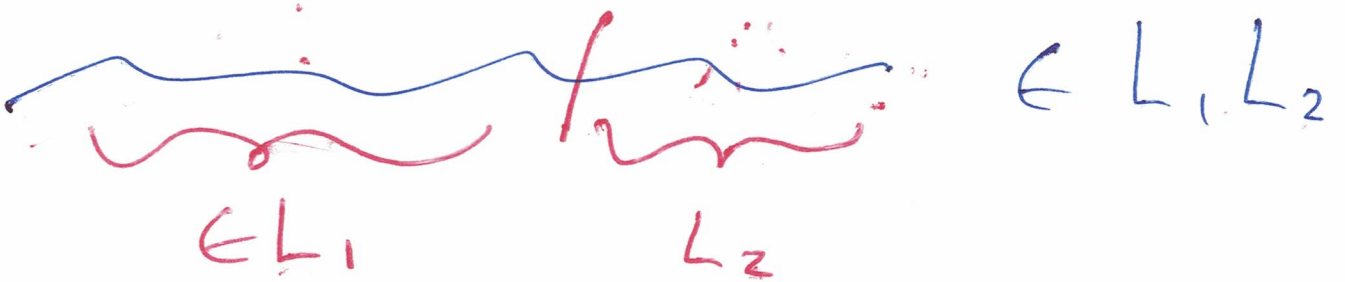
(4.1). Appear($L_1 + L_2$, b) is reg. why?

Appear($L_1 + L_2$, b) = Appear(L_1, b) +

Appear(L_2, b) is reg

Since Appear(L_1, b) and Appear(L_2, b) are reg and regularity is closed under "+".

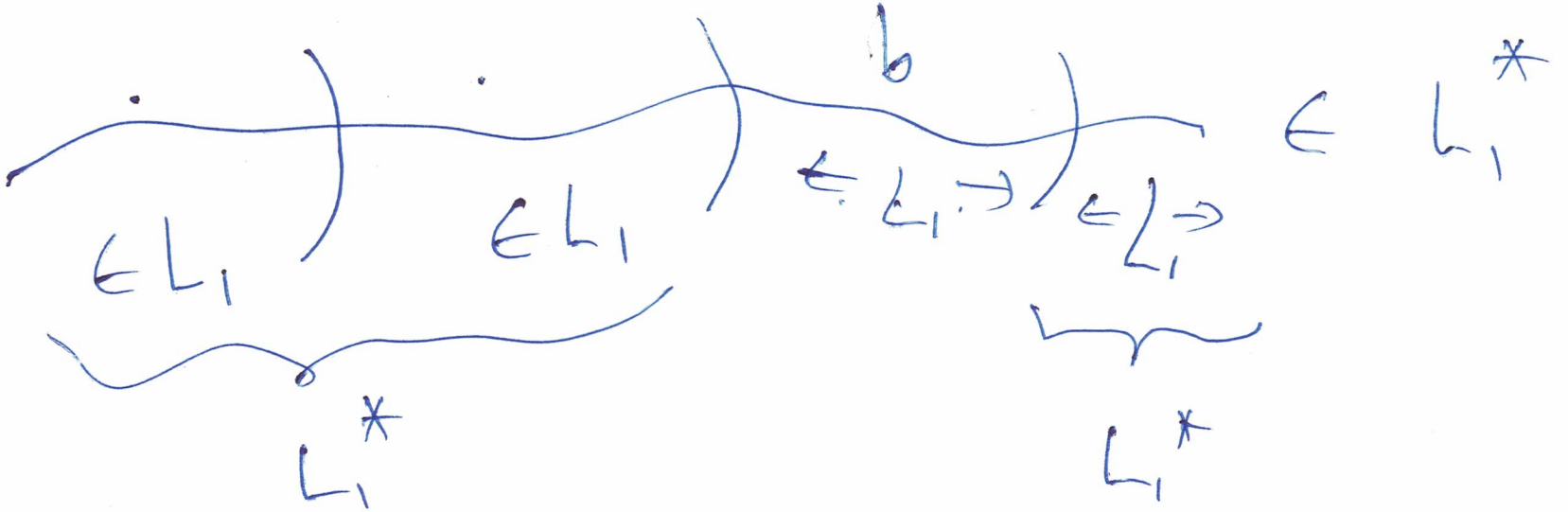
(4.2). Appear$(L_1 L_2, b)$ is reg. why?



$$\text{Appear}(L_1 L_2, b) = \text{Appear}(L_1, b) \cdot L_2 + L_1 \cdot \text{Appear}(L_2, b)$$

is regular since L_1, L_2, Appear(L_1, b), Appear(L_2, b) are reg and regularity is closed under '+' and '·'!

(4.3). Appear(L_1^*, b) is reg, why?



$$\text{Appear}(L_1^*, b) = L_1^* \cdot \text{Appear}(L_1, b) \cdot L_1^*$$

is regular since L_1 and Appear(L_1, b) are reg and regularity is closed under '*' and '·'.